

## CO2-AT2

### Scenario Based Assessment

#### SOURCE CODE:

```
#include <stdio.h>

#include <math.h>

int main()
{
    int n, i, j;

    float x[100], y[100];

    float minDist, dist;

    int p1 = 0, p2 = 1;

    printf("Enter number of warehouses: ");

    scanf("%d", &n);

    printf("Enter coordinates (x y):\n");

    for(i=0;i<n;i++)
    {
        scanf("%f %f",&x[i],&y[i]);
    }

    minDist = sqrt(pow(x[0]-x[1],2)+pow(y[0]-y[1],2));

    for(i=0;i<n-1;i++)
    {
        for(j=i+1;j<n;j++)
        {
            dist = sqrt(pow(x[i]-x[j],2)+pow(y[i]-y[j],2));

            if(dist < minDist)
            {
                minDist = dist;
```

```
        p1 = i;
        p2 = j;
    }
}
}
printf("\nClosest Warehouses:\n");
printf("Warehouse %d and Warehouse %d\n", p1+1, p2+1);
printf("Minimum Distance = %.2f\n", minDist);
return 0;
}
```